

The AWS AI and ML Lifecycle



The four MLA-C02 domains.

-Sania Sadaf

MLA-C02 gives nearly equal weight to the full lifecycle



Domain 1 turns raw information into trustworthy model input

01 Collect

S3, databases, Kinesis, Kafka, and other sources provide batch or streaming data.

02 Transform

Glue, EMR, Lambda, and SageMaker tools clean and reshape information.

03 Engineer

Features, embeddings, chunking, tokenization, and metadata create useful representations.

04 Validate

Quality checks, labeling, privacy controls, and bias management improve trust.

Domain 2 develops both predictive models and generative AI

Amazon SageMaker AI

Supports data science workflows, algorithms, training jobs, tuning, experiments, evaluation, and model management.

Amazon Bedrock

Provides managed access to foundation models, prompt tools, evaluations, knowledge bases, guardrails, and agents.

Domain 3 converts a model into a repeatable service

01

02

03

04

Package

Prepare artifacts, containers, prompts, or imported models.

Provision

Select compute, networking, storage, and scaling capacity.

Deploy

Choose real-time, serverless, asynchronous, or batch inference.

Automate

Use pipelines, testing, registries, approvals, and rollback strategies.

Domain 4 keeps production ML controlled

01 Monitor

Track model quality, drift, errors, latency, throughput, agent behavior, and infrastructure.

02 Maintain

Retrain, update, roll back, and manage versions when conditions change.

03 Optimize

Balance capacity, purchasing options, token usage, vector storage, and performance.

04 Secure

Apply least privilege, encryption, network controls, logging, governance, and compliance.

Thank you :)